

SECURE SOFTWARE ENGINEERING CSCE469 SEC 001 Fall 2025

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Course Information

Professor: [Hamid Bagheri \(http://www.cse.unl.edu/~hbagheri\)](http://www.cse.unl.edu/~hbagheri) and David Hemsath

Office: Avery Hall 365

Office Hours: Tuesday 10:45-11:45 AM and by appointment.

Email: bagheri@unl.edu, (<mailto:bagheri@unl.edu>), dhemsath3@unl.edu (<mailto:dhemsath@gmail.com>)

Class Hours: Tuesday Thursday 9:30AM - 10:45AM

Classroom: Kiewit Hall (HAH) A249

TA:

Office: Avery Hall 104D

Email:

Office Hours:

Prerequisite: Software Engineering.

Credit: 3 credit hours

Handbook of Software Engineering, Springer 2019, ISBN 978-3-030-00261-9

Text: Janca, Tanya. *Alice and Bob Learn Secure Coding*. Wiley, 2025. ISBN numbers: 9781394171705 (paperback), 9781394171729 (ePDF), 9781394171712 (ePub).

Grading Criteria

- Quizzes, homework, assignments - 35 %
- Exams - 30 %
- Project - 35 %

We will assign readings from the course textbooks, and additional readings from other sources. Exams will address all material covered in class, assigned in readings, or covered in assignments. We do not grant exemptions from exams, and do not allow make-up exams. The course will consist of some quizzes and presentations as well as a term project. The course will have a substantial project which requires an oral presentation of the initial ideas (proposal) and the final results, among other things. The course will require strong class participation. You are expected to read the material assigned and participate in discussions.

We will attempt to return deliverables with grades as quickly as possible, consistent with our ability to do a thorough job and provide sufficient feedback.

Upon receiving graded deliverables, if you believe that errors have been made in grading your deliverables, you may see us to discuss

your concerns. However, changes in scoring will be considered only in cases in which: 1) we made a numerical error in tallying scores; 2) we misinterpreted your answer or solution and your answer or solution as given is actually appropriate as can be seen by looking at that answer or solution itself; 3) the problem statement was provably ambiguous and you correctly addressed a plausible alternative interpretation of the statement. Note: you have one week following return of graded items in which to discuss any such scoring-related problems with the graders. We are happy, however, to discuss non-scoring-related matters such as ways in which you might have improved your answers or solutions at any time.

Grade Assignments are made based on the following:

A: 93-100 A-: 90-92

B+: 87-89 B: 83-86 B-: 80-82

C+: 77-79 C: 73-76 C-: 70-72

D+: 67-69 D: 63-66 D-: 60-62 F: 0-59

Course, Department, and University Policies and Resources

Academic Integrity Policy:

All homework assignments, exams, etc. must be your own work. No direct collaboration with fellow students, past or current, is allowed *unless otherwise stated*. The Computer Science and Engineering department has an Academic Integrity Policy. [_\(http://cse.unl.edu/ugrad/resources/academic_integrity.php.html\)](http://cse.unl.edu/ugrad/resources/academic_integrity.php.html). All students enrolled in any computer science course are bound by this policy. You are expected to read, understand, and follow this policy. Violations will be dealt with on a case by case basis and may result in a failing assignment or a failing grade for the course itself.

Accommodation:

Students with disabilities are encouraged to contact the instructor for a confidential discussion of their individual needs for academic accommodation. It is the policy of the University of Nebraska-Lincoln to provide flexible and individualized accommodations to students with documented disabilities that may affect their ability to fully participate in course activities or to meet course requirements. To receive accommodation services, students must be registered with the Services for Students with Disabilities (SSD) office, 132 Canfield Administration, 472-3787 voice or TTY.

Attendance:

We will cover some material in class that is not covered in the text; your assignments and test questions may be based on that material. In particular, many of the details necessary to complete assignments will be presented in class. We also have some in-class assignments and quizzes. Thus, it is to your benefit to attend class. Because late arrivals are distracting, we also ask that you arrive at class on time.

Class Preparation and Participation:

We expect you to be prepared for class, participate in discussions, and ask and answer questions.

Student Resource Center:

The School of Computing maintains a Student Resource Center [_\(http://cse.unl.edu/src/index.html\)_](http://cse.unl.edu/src/index.html) in Avery Hall 13A.

Email Responsibilities:

It is school of computing policy that all students in SoC courses are expected to regularly check their email so that they do not miss important announcements.

Departmental Suggestion Box:

The school of computing has an [Anonymous Online Suggestion Box](http://cse.unl.edu/department/suggestion.php.html) (<http://cse.unl.edu/department/suggestion.php.html>) that you may use to voice your concerns about any problems in the course or department if you do not wish to be identified.

Detailed Course Schedule

The schedule below is tentative and subject to change. Please check the course web page frequently during the semester.

Class	Day/Date	Topic	Handouts	Assignments	
1	Tue 08/26	https://canvas.unl.edu/courses/201137/files/22166439?wrap=1 Course Overview		M01_Listening_and_Reading_Assignment	
2	Thu 08/28	M01A: An Introduction to Information Security & Privacy for Secure Software Engineering			
3	Tue 09/02	M01B: Introductory Security & Privacy Fundamentals		M02_Reading_Assignment	

4	Thu 09/04	Project Overview	Project Proposal Guidelines	Team members, due Tuesday 09/09, at noon Project proposal due Thursday 10/02, at 11:59 pm. Final Report and Artifacts due Tuesday 12/02, at 11:59 pm	
5	Tue 09/09	Analysis-Introduction (1)			
6	Thu 09/11	Analysis-Introduction (2)			

7	Tue 09/16	M02A: More Fundamentals			
8	Thu 09/18	M02B: More Fundamentals		M03_Reading_Assignment	
9	Tue 09/23	Android-Introduction (1)			
10	Thu 09/25	Android-Introduction (2)			
11	Tue 09/30	M03: Improving		M04_Reading_Assignment	
12	Thu 10/02	M04: Achieving		M08_Reading_Assignment	
13	Tue 10/07	Project Proposal Presentations (1)			
14	Thu 10/09	Project Proposal Presentations (2)			
15	Tue 10/14	M08: Vulnerabilities		M09-M12_Reading_Assignment	
16	Thu 10/16	MIDTERM EXAM (80		Closed book, closed notes, taken in	

		MINUTES)		person (not remotely) using the Canvas LMS. The number of questions, etc. will be matched to the exam length.	
17	Tue 10/21	***NO CLASS***		Fall semester break	
18	Thu 10/23	Android Security Analysis			
19	Tue 10/28	Mitigation Example			
20	Thu 10/30	Exhaustive Analysis (prevention example)			
21	Tue 11/04	M09A: Secure SDLC			
22	Thu 11/06	M09B: Secure SDLC			
23	Tue 11/11	M09C: Secure SDLC			
24	Thu 11/13	M09D: Secure SDLC			
25	Tue 11/18	Static Analysis and Dynamic Analysis			

26	Thu 11/20	Formal Methods and Security Analysis Taxonomy			
27	Tue 11/25				
28	Thu 11/27	***NO CLASS***		U.S. Thanksgiving holiday	
29	Tue 12/02				
30	Thu 12/04	Final Project Presentations (1)			
31	Tue 12/09	Final Project Presentations (2)			
32	Thu 12/11	Final Project Presentations (3)			
	TBA	FINAL EXAM (80 minutes) HH:MM-		Closed book, closed notes, taken in person (not remotely) using the Canvas	

		HH:MM, Kiewit Hall Rm A249		LMS. The number of questions, etc. will be matched to the exam length.	
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Course Summary:

Date	Details	Due
Thu Aug 28, 2025	 M01 Listening and Reading Assignment https://canvas.unl.edu/courses/201137/assignments/2046213	due by 9:30am
Tue Sep 16, 2025	 M02 Reading Assignment (https://canvas.unl.edu/courses/201137/assignments/2110614)	due by 9:30am
Tue Sep 30, 2025	 M03 Reading Assignment (https://canvas.unl.edu/courses/201137/assignments/2110620)	due by 9:30am
Thu Oct 2, 2025	 M04 Reading Assignment (https://canvas.unl.edu/courses/201137/assignments/2110621)	due by 9:30am
Tue Oct 14, 2025	 M08 Reading Assignment (https://canvas.unl.edu/courses/201137/assignments/2114463)	due by 9:30am

Date	Details	Due
Thu Nov 13, 2025	 M09-M12 Reading Assignment (https://canvas.unl.edu/courses/201137/assignments/2114484)	due by 9:30am